

Basic Properties and Contraction of AFEM

Adaptive Finite Element Methods for Elliptic PDEs — Lecture 7

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github.com/morinpedro/afem-cimpa-Rosario-2026

Convergence of SOLVE $\rightarrow$ ESTIMATE $\rightarrow$ MARK $\rightarrow$ REFINES

We consider two scenarios

• **Piecewise constant data and interior node property:** We assume that

- The coefficient $A(x)$ and the forcing term $f(x)$ are constant on each element of the coarsest initial mesh $\mathcal{T}_0$.
- Each marked element is refined in such a way that an interior node is created inside it.

$\rightsquigarrow$ Contraction of the error $\|u - U_k\|$.

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• **General case:** We assume that $A(x)$ and $f(x)$ are piecewise smooth and do not enforce the interior node property. We prove

Contraction of the *quasi error* $\zeta_k = \left(\|u - U_k\|^2 + \gamma \mathcal{E}_k^2 \right)^2$.

Roadmap

- 1 The adaptive loop: notation
- 2 Four basic properties
- 3 Contraction: vanishing oscillation
- 4 Estimator reduction
- 5 Contraction: general data
- 6 Synthesis
- 7 Localized upper bound

The adaptive loop: notation

Nested meshes from AFEM

- **Nested meshes:** The loop of Lectures 5–6 produces a sequence of nested conforming meshes

$$\mathcal{T}_0 \leq \mathcal{T}_1 \leq \mathcal{T}_2 \leq \cdots, \quad \mathbb{V}_k := \mathbb{V}_{\mathcal{T}_k} \subset \mathbb{V}_{k+1},$$

with Galerkin solutions $U_k \in \mathbb{V}_k$, indicators $\mathcal{E}_k(U_k, T)$, and marked sets $\mathcal{M}_k \subset \mathcal{T}_k$ chosen by Dörfler marking.

- **Nestedness is automatic:** bisection never coarsens, so $\mathbb{V}_k \subset \mathbb{V}_{k+1}$ always.
- **Central object:** The single-step increment $U_{k+1} - U_k \in \mathbb{V}_{k+1}$ is central below: it appears in both the localized upper bound and in the estimator reduction.

Abbreviations used throughout

$$e_k = \|u - U_k\|_{\Omega}, \quad E_k = \|U_{k+1} - U_k\|_{\Omega}, \quad \mathcal{E}_k = \mathcal{E}_k(U_k, \mathcal{T}_k), \quad \text{osc}_k = \text{osc}_k(U_k).$$

Four basic properties

Four basic properties

Specialized to nested meshes $\mathcal{T}_k \leq \mathcal{T}_{k+1}$:

- ❶ **Upper bound** (reliability, Lecture 5): $e_k^2 \leq C_1 \mathcal{E}_k^2$.
- ❷ **Lower bound** (efficiency, Lecture 5): $C_2 \mathcal{E}_k^2 \leq e_k^2 + \text{osc}_k^2$.

- ❸ **Discrete lower bound** (thanks to interior node property, Lecture 5):

$$C_2 \mathcal{E}_k^2(U_k, T) \leq \|U_{k+1} - U_k\|_{\omega_k(T)}^2 + \text{osc}_k^2(U_k, \omega_k(T)).$$

- ❹ **Pythagoras orthogonality:** $e_k^2 = e_{k+1}^2 + E_k^2$,

because $\mathcal{B}[u - U_{k+1}, U_{k+1} - U_k] = 0$ for $U_{k+1} - U_k \in \mathbb{V}_{k+1}$.

Contraction: vanishing oscillation

The easy case: no oscillation

- **Piecewise constant data:** When $f, A \in \mathcal{P}_0(\mathcal{T}_0)$ (piecewise constant on the *initial* mesh), $\text{osc}_k(U_k) \equiv 0$ on every refinement, and the estimator is equivalent to the energy error with **no correction**.
- **Direct contraction:** We can then contract e_k *directly*, converting Dörfler marking into an energy decrement via the interior node property.

Contraction theorem (no oscillation)

Theorem 1 (contraction, piecewise constant data)

Let $\text{osc}_k(U_k) = 0$ and $\mathcal{M}_k$ satisfy Dörfler marking $\mathcal{E}_k(U_k, \mathcal{M}_k) \geq \theta \mathcal{E}_k(U_k, \mathcal{T}_k)$, $0 < \theta < 1$. If every $T \in \mathcal{M}_k$ gets an interior node of $\mathcal{T}_{k+1}$, then

$$e_{k+1} \leq \alpha e_k, \quad \alpha = \left(1 - \theta^2 \frac{C_2}{C_1}\right)^{1/2} < 1.$$

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$$e_{k+1} \leq \alpha e_k, \quad \alpha = \left(1 - \theta^2 \frac{C_2}{C_1}\right)^{1/2} < 1.$$

- **Proof:** By Pythagoras, $e_{k+1}^2 = e_k^2 - E_k^2$; bound E_k^2 below via the interior node property, Dörfler marking, and the upper bound, in this order:

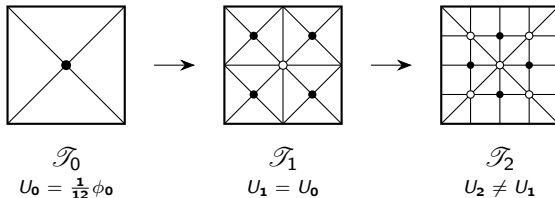
$$E_k^2 \geq C_2 \mathcal{E}_k^2(U_k, \mathcal{M}_k) \geq \theta^2 C_2 \mathcal{E}_k^2 \geq \theta^2 \frac{C_2}{C_1} e_k^2. \quad \square$$

- **Linear convergence:** Iterating: **linear convergence** $e_k \leq \alpha^k e_0 \rightarrow 0$.

Why the interior node property is necessary

Counterexample (Morin–Nochetto–Siebert)

- $A = I$, $f = 1$, $\Omega = (0, 1)^2$, criss-cross initial mesh $\mathcal{T}_0$.
- Two bisection steps adding *no* interior node leaves the Galerkin solution *unchanged*: $U_1 = U_0 = \frac{1}{12}\phi_0$, so
- $E_0 = 0$ and $e_0 = e_1$ — **no contraction**.
- Only $\mathcal{T}_2$, which finally introduces an interior node, gives $U_2 \neq U_1$.



Estimator reduction

Setting: the estimator across a refinement step

Estimator recap (Lecture 5):

$$\mathcal{E}_{\mathcal{T}}(U) = \left(\sum_{T \in \mathcal{T}} \mathcal{E}_{\mathcal{T}}^2(U, T) \right)^{1/2}, \quad \mathcal{E}_{\mathcal{T}}^2(U, T) = h_T^2 \|r\|_{L^2(T)}^2 + h_T \|j\|_{L^2(\partial T)}^2.$$

Two competing effects between $\mathcal{T}_k$ and $\mathcal{T}_{k+1}$

- New elements are **smaller**: this shrinks the h -weighted indicators on refined elements.
- The discrete solution changes from U_k to U_{k+1} : this **perturbs** every indicator, but only in a controlled, Lipschitz way.

- **Plan:** We quantify each effect, then combine them.

Lipschitz dependence on the discrete argument

Lemma 1 (Lipschitz dependence)

For $W_1, W_2 \in \mathbb{V}_{\mathcal{T}}(\mathcal{T})$ and each $T \in \mathcal{T}$,

$$|\mathcal{E}_{\mathcal{T}}(W_1, T) - \mathcal{E}_{\mathcal{T}}(W_2, T)| \lesssim \|\nabla(W_1 - W_2)\|_{L^2(\omega_T)}.$$

Therefore $|\mathcal{E}_{\mathcal{T}}(W_1) - \mathcal{E}_{\mathcal{T}}(W_2)| \leq \sqrt{C_0} \|W_1 - W_2\|_{\Omega}.$

Sketch of proof: Recall that $\mathcal{E}_{\mathcal{T}}(W, T)^2 = h_T^2 \|\operatorname{div}(A \nabla W) + f\|_{L^2(T)}^2 + h_T \| \llbracket A \nabla W \rrbracket \|_{L^2(\partial T)}^2$

Let $W_1, W_2 \in \mathbb{V}_{\mathcal{T}}(\mathcal{T})$, and $T \in \mathcal{T}$. For the interior residual we proceed as follows:

$$\begin{aligned} \left| h_T \|f + \operatorname{div}(A \nabla W_1)\|_{L^2(T)} - h_T \|f + \operatorname{div}(A \nabla W_2)\|_{L^2(T)} \right| &\leq h_T \|\operatorname{div}(A \nabla W_1) - \operatorname{div}(A \nabla W_2)\|_{L^2(T)} \\ &= h_T \|\operatorname{div}(A \nabla(W_1 - W_2))\|_{L^2(T)}. \end{aligned}$$

Let $W = W_1 - W_2$

$$\begin{aligned} h_T \|\operatorname{div}(A \nabla(W))\|_{L^2(T)} &\leq h_T \|\operatorname{div}(A)\|_{L^\infty(T)} \|\nabla W\|_{L^2(T)} + h_T \|A\|_{L^\infty(T)} \|D^2 W\|_{L^2(T)} \\ &\lesssim h_T \|\nabla W\|_{L^2(T)} + h_T h_T^{-1} \|\nabla W\|_{L^2(T)} \lesssim \|\nabla W\|_{L^2(T)} = \|\nabla(W_1 - W_2)\|_{L^2(T)}. \end{aligned}$$

Estimator reduction: statement

Proposition 1 (estimator reduction)

Let $\mathcal{T}_{k+1} \geq \mathcal{T}_k$ bisect every marked $T \in \mathcal{M}_k$ at least once, and set $\lambda := 1 - 2^{-1/d} \in (0, 1)$. There is $C_0 > 0$ (shape regularity and data only) such that for every $\delta > 0$,

$$\mathcal{E}_{k+1}^2(U_{k+1}) \leq (1 + \delta) \left[\mathcal{E}_k^2(U_k) - \lambda \mathcal{E}_k^2(U_k, \mathcal{M}_k) \right] + (1 + \delta^{-1}) C_0 E_k^2.$$

Reading it

The bracket says: refining $\mathcal{M}_k$ *shrinks* the estimator by a fixed fraction λ of its marked part. The last term is the price of the solution having changed — and it is controlled, via the localized upper bound, by the very same marked set.

Proof — Step 1: reduction for a *fixed* function

- Fixed function:** Fix $V \in \mathbb{V}_k \subset \mathbb{V}_{k+1}$. Bisecting T once halves $|T|$, so $h_{T'} \leq 2^{-1/d} h_T$ for a descendant $T' \subset T$; since V stays polynomial on descendants and interior edges created *inside* T carry no jump,

$$\sum_{T' \subset T} \mathcal{E}_{k+1}^2(V, T') \leq 2^{-1/d} \mathcal{E}_k^2(V, T).$$

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$$\sum_{T' \subset T} \mathcal{E}_{k+1}^2(V, T') \leq 2^{-1/d} \mathcal{E}_k^2(V, T).$$

- Summation:** Unrefined elements keep their indicator unchanged. Summing over $\mathcal{T}_k$ and isolating the marked (hence refined) elements $\mathcal{M}_k \subset \mathcal{R}_k = \mathcal{T}_k \setminus \mathcal{T}_{k+1}$:

$$\mathcal{E}_{k+1}^2(V) \leq \underbrace{\mathcal{E}_k^2(V) - (1 - 2^{-1/d})}_{\lambda} \sum_{T \in \mathcal{R}_k} \mathcal{E}_k^2(V, T) \leq \mathcal{E}_k^2(V) - \lambda \mathcal{E}_k^2(V, \mathcal{M}_k).$$

Proof — Step 2: combine with the Lipschitz bound

- **Young's inequality:** $(a + b)^2 \leq (1 + \delta)a^2 + (1 + \delta^{-1})b^2$, applied to

$$\mathcal{E}_{k+1}(U_{k+1}) = \mathcal{E}_{k+1}(U_k) + (\mathcal{E}_{k+1}(U_{k+1}) - \mathcal{E}_{k+1}(U_k)):$$

$$\mathcal{E}_{k+1}^2(U_{k+1}) \leq (\mathcal{E}_{k+1}(U_k) + \sqrt{C_0} E_k)^2 \leq (1 + \delta) \mathcal{E}_{k+1}^2(U_k) + (1 + \delta^{-1}) C_0 E_k^2,$$

using the Lipschitz Lemma with $W_1 = U_{k+1}$, $W_2 = U_k$ on $\mathcal{T}_{k+1}$.

- **Apply Step 1:** Now apply Step 1 with $V = U_k$ to $\mathcal{E}_{k+1}^2(U_k)$:

$$\mathcal{E}_{k+1}^2(U_{k+1}) \leq (1 + \delta) \left[\mathcal{E}_k^2(U_k) - \lambda \mathcal{E}_k^2(U_k, \mathcal{M}_k) \right] + (1 + \delta^{-1}) C_0 E_k^2. \quad \square$$

Estimator reduction is the engine of the *general* contraction proof, next.

Contraction: general data

Estimator reduction: review

Proposition 2 (estimator reduction)

Let $\mathcal{T}_{k+1} \geq \mathcal{T}_k$ bisect every marked $T \in \mathcal{M}_k$ at least once, and set $\lambda := 1 - 2^{-1/d} \in (0, 1)$. There is $C_0 > 0$ (shape regularity and data only) such that for every $\delta > 0$,

$$\mathcal{E}_{k+1}^2(U_{k+1}) \leq (1 + \delta) \left[\mathcal{E}_k^2(U_k) - \lambda \mathcal{E}_k^2(U_k, \mathcal{M}_k) \right] + (1 + \delta^{-1}) C_0 E_k^2.$$

If we combine with Dörfler's Marking Strategy

$$\mathcal{E}_k^2(U_k, \mathcal{M}_k) \geq \theta \mathcal{E}_k^2(U_k) \quad \implies \quad \mathcal{E}_k^2(U_k) - \lambda \mathcal{E}_k^2(U_k, \mathcal{M}_k) \leq \mathcal{E}_k^2(U_k) - \lambda \theta^2 \mathcal{E}_k^2(U_k).$$

Then

$$\mathcal{E}_{k+1}^2(U_{k+1}) \leq (1 + \delta)(1 - \lambda \theta^2) \mathcal{E}_k^2(U_k) + (1 + \delta^{-1}) C_0 E_k^2.$$

Why energy alone can stall

- **Energy stalls:** In the counterexample, $U_{k+1} = U_k$ so $E_k = 0$ and $e_{k+1} = e_k$: the energy error **stalls**.
- **Estimator still drops:** Yet with $E_k = 0$, estimator reduction gives

$$\mathcal{E}_{k+1}^2(U_{k+1}) \leq (1 + \delta)(1 - \lambda\theta^2) \mathcal{E}_k^2(U_k),$$

which *strictly decreases* for small $\delta > 0$.

- **Neither alone works:** Neither e_k nor $\mathcal{E}_k(U_k)$ contracts *alone* in general — energy can stall while the estimator drops, and the estimator is not even monotone on its own (it depends on the changing U_k).

The quasi-error

The fix: a weighted combination

$$\zeta_k^2 := e_k^2 + \gamma \mathcal{E}_k^2 \quad (\gamma > 0 \text{ to be chosen}).$$

- **Blend of two worlds:** This blends a *continuous* quantity (e_k) with a *discrete* one ($\mathcal{E}_k$), neither of which suffices alone — the hallmark of the Cascón–Kreuzer–Nochetto–Siebert convergence theory.
- **Goal:** We show ζ_k contracts, for *any* data, with or without oscillation, with or without the interior node property.

Contraction of the quasi-error: statement

Theorem 2 (contraction of the quasi-error)

Let $0 < \theta \leq 1$, Dörfler marking with parameter θ , Refine bisecting every $T \in \mathcal{M}_k$. There are $\gamma > 0$, $\alpha \in (0, 1)$ (shape regularity, data, θ only) with

$$\zeta_{k+1}^2 = e_{k+1}^2 + \gamma \mathcal{E}_{k+1}^2 \leq \alpha^2 \zeta_k^2 = \alpha^2 (e_k^2 + \gamma \mathcal{E}_k^2).$$

Consequence

$$\zeta_k \leq \alpha^k \zeta_0, \quad \rightsquigarrow \quad \text{linear convergence of the quasi-error.}$$

Hence,

$$e_k \leq \zeta_k \leq \alpha^k \zeta_0 \quad \rightsquigarrow \quad \text{AFEM converges linearly, unconditionally.}$$

Proof — Steps 1–2: assemble and choose δ, γ

- **Pythagoras:**
$$e_{k+1}^2 = e_k^2 - E_k^2.$$

- **Estimator reduction and Dörfler marking:** (any $\delta > 0$)

$$\mathcal{E}_{k+1}^2 \leq (1 + \delta)(1 - \lambda\theta^2)\mathcal{E}_k^2 + (1 + \delta^{-1})C_0 E_k^2.$$

- **Combine:** Multiply the second by γ and add the first:

$$\zeta_{k+1}^2 \leq e_k^2 + (\gamma(1 + \delta^{-1})C_0 - 1)E_k^2 + \gamma(1 + \delta)(1 - \lambda\theta^2)\mathcal{E}_k^2.$$

- **Choose δ :** Fix δ by $(1 + \delta)(1 - \lambda\theta^2) = 1 - \frac{1}{2}\lambda\theta^2$, i.e. $\delta = \frac{\lambda\theta^2}{2(1 - \lambda\theta^2)} > 0$.

- **Choose γ :** then $\gamma > 0$ by $\gamma(1 + \delta^{-1})C_0 = 1$ — this **exactly cancels** the E_k^2 term:

$$\zeta_{k+1}^2 \leq e_k^2 + \gamma(1 - \frac{1}{2}\lambda\theta^2)\mathcal{E}_k^2.$$

Proof — Step 3: Reliability

So far,

$$\zeta_{k+1}^2 \leq e_k^2 + \gamma \left(1 - \frac{\lambda\theta^2}{2}\right) \mathcal{E}_k^2.$$

• **Reliability:** $e_k^2 \leq C_1 \mathcal{E}_k^2$:

$$\begin{aligned} \zeta_{k+1}^2 &\leq \left(1 - \frac{\gamma\lambda\theta^2}{4C_1}\right) e_k^2 + \frac{\gamma\lambda\theta^2}{4C_1} e_k^2 + \gamma \left(1 - \frac{1}{2}\lambda\theta^2\right) \mathcal{E}_k^2 \\ &\leq \left(1 - \frac{\gamma\lambda\theta^2}{4C_1}\right) e_k^2 + \frac{\gamma\lambda\theta^2}{4} \mathcal{E}_k^2 + \gamma \left(1 - \frac{1}{2}\lambda\theta^2\right) \mathcal{E}_k^2 \\ &\leq \left(1 - \frac{\gamma\lambda\theta^2}{4C_1}\right) e_k^2 + \left(1 - \frac{\lambda\theta^2}{4}\right) \gamma \mathcal{E}_k^2 \leq \alpha^2 (e_k^2 + \gamma \mathcal{E}_k^2) \zeta_k^2 = \alpha^2 \zeta_k^2 \end{aligned}$$

with $\alpha^2 := \max \left\{ 1 - \frac{\gamma\lambda\theta^2}{4C_1}, 1 - \frac{\lambda\theta^2}{4} \right\} < 1$.

□

Synthesis

Ingredients of the two proofs

- **Vanishing oscillation:** contracts e_k *itself*, but needs $\text{osc}_k(U_k) = 0$ *and* the interior node property.
- **General case:** contracts the quasi-error $\zeta_k^2 = e_k^2 + \gamma \mathcal{E}_k^2$, needs *neither*.

The four ingredients of the general proof

- Dörfler marking.
- Pythagoras orthogonality.
- The a posteriori **upper** bound (reliability).
- Estimator reduction.

Notably *not* the lower bound (efficiency) — that returns only in the convergence-*rate* analysis ahead.

Quasi-error vs. total error

- **Total error:** Reliability and efficiency together give the *total error* equivalence

$$\mathcal{E}_k^2 \simeq e_k^2 + \text{osc}_k^2 =: \text{total error.}$$

Two equivalent quantities, two different jobs

The quasi-error $\zeta_k^2 = e_k^2 + \gamma \mathcal{E}_k^2$ is *equivalent* to the total error, and it is the total error that governs the optimal decay *rates* ahead. So efficiency, unused here, is exactly what ties the contracted quantity to approximation classes.

Mark is driven by $\mathcal{E}_k$ *alone*: a single Dörfler marking suffices for contraction between consecutive steps.

Four Five basic properties

❶ **Upper bound** (reliability, Lecture 5): $e_k^2 \leq C_1 \mathcal{E}_k^2$.

❷ **Lower bound** (efficiency, Lecture 5): $C_2 \mathcal{E}_k^2 \leq e_k^2 + \text{osc}_k^2$.

❸ **Discrete lower bound** (thanks to interior node property, Lecture 5):

$$C_2 \mathcal{E}_k^2(U_k, T) \leq \|U_{k+1} - U_k\|_{\omega_k(T)}^2 + \text{osc}_k^2(U_k, \omega_k(T)).$$

❹ **Pythagoras orthogonality:** Since $\mathcal{B}[u - U_{k+1}, U_{k+1} - U_k] = 0$ ($U_{k+1} - U_k \in \mathbb{V}_{k+1}$)

$$e_k^2 = e_{k+1}^2 + E_k^2.$$

❺ **Localized upper bound** $\mathcal{R}_k^+$: discrete neighborhood of the refined set $\mathcal{R}_k = \mathcal{I}_k \setminus \mathcal{I}_{k+1}$

$$E_k^2 \leq C_1 \mathcal{E}_k^2(U_k, \mathcal{R}_k^+).$$

Localized upper bound

Localized upper bound: statement

- **No local bound continuous-to-discrete:** Between the *continuous* and discrete solution, no local upper bound is possible. Between **two discrete** solutions on nested meshes, it is:

Lemma 2 (localized upper bound)

Let $\mathcal{T}_{k+1} \geq \mathcal{T}_k$ with refined set $\mathcal{R}_k = \mathcal{T}_k \setminus \mathcal{T}_{k+1}$, and $\mathcal{R}_k^+$ its discrete $\mathcal{T}_k$ -neighborhood. Then

$$E_k = \|U_{k+1} - U_k\|_{\Omega} \leq \sqrt{C_1} \mathcal{E}_k(U_k, \mathcal{R}_k^+), \quad \mathcal{E}_k(U_k, \mathcal{R}_k^+) = \left(\sum_{T \in \mathcal{R}_k^+} \mathcal{E}_k^2(U_k, T) \right)^{1/2}.$$

- **Interpretation:** Only the elements *actually refined* (and their neighbors) control how much the solution can change — unrefined regions contribute nothing to E_k .

Localized upper bound: proof

- Setup:** Set $V = U_{k+1} - U_k \in \mathbb{V}_{k+1}$, and let $I_k V \in \mathbb{V}_k$ be its quasi-interpolant onto the coarse space. Galerkin orthogonality on both meshes, with residual $\langle R_k, w \rangle = \langle f, w \rangle - \mathcal{B}[U_k, w]$:

$$\|U_{k+1} - U_k\|_{\Omega}^2 = \mathcal{B}[U_{k+1} - U_k, V] = \langle f, V \rangle - \mathcal{B}[U_k, V] = \langle R_k, V \rangle = \langle R_k, V - I_k V \rangle,$$

the last step because $\langle R_k, I_k V \rangle = 0$.

The key localization

$I_k V$ reproduces V exactly wherever refinement has not touched the discrete neighborhood, so $V - I_k V$ is supported only on $\bigcup_{T \in \mathcal{R}_k} \omega_{\mathcal{T}_k}(T) = \bigcup_{T \in \mathcal{R}_k^+} T$.

- Localized reliability:** Repeating the reliability argument of Lecture 5, but summing only over $\mathcal{R}_k^+$:

$$\langle R_k, V - I_k V \rangle = \sum_{T \in \mathcal{R}_k^+} \int_T r(V - I_k V) + \int_{\partial T} j(V - I_k V) \leq C_1 \mathcal{E}_k(U_k, \mathcal{R}_k^+) \|V\|_{\Omega}.$$

Towards optimality: Four basic properties

① **Upper bound** (reliability, Lecture 5): $e_k^2 \leq C_1 \mathcal{E}_k^2$.

② **Lower bound** (efficiency, Lecture 5): $C_2 \mathcal{E}_k^2 \leq e_k^2 + \text{osc}_k^2$.

③ **Pythagoras orthogonality**: Since $\mathcal{B}[u - U_{k+1}, U_{k+1} - U_k] = 0$ ($U_{k+1} - U_k \in \mathbb{V}_{k+1}$)

$$e_k^2 = e_{k+1}^2 + E_k^2.$$

④ **Localized upper bound** $\mathcal{R}_k^+$: discrete neighborhood of the refined set $\mathcal{R}_k = \mathcal{T}_k \setminus \mathcal{T}_{k+1}$

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Towards optimality: Four basic properties

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Next: Lecture 8 — quasi-optimal convergence rates.